FINITE PRESENTATION SPACES GLUE ALONG ZARISKI OPENS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4164-4195, following the statement at 4153-4162.

Editorial note. No omissions; the note that no cocycle data are needed is justified in the text.

Allowed standard prerequisites (editorial).

- Gluing sheaves on a covering. Sites, Lemma sites-lemma-mapping-property-glue (cross-chapter, cited at ~4176). Sheaves on (Sch/S)_{fppf} are triples (F_U, F_W, iso) for the Zariski covering.
- Glueing algebraic spaces. Algebraic Spaces, Lemma spaces-lemma-glueing-algebraic-spaces (cross-chapter, cited at ~4184-4187). F is an algebraic space iff F_U and F_W are.
- Local-on-base criteria for finite presentation. Morphisms of Spaces, Lemmas spaces-morphisms-lemma-quasi-compact-local, spaces-morphisms-lemma-separated-local, spaces-morphisms-lemma-finite-presentation-local (cross-chapter, cited at ~4191-4194). fp over S iff fp over U and fp over W.

Source reference. Original reference 04TS: focused-seeds/geometry/S9-spaces-limits-relative-glueing/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02WR: focused-seeds/geometry/S9-spaces-limits-relative-glueing/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0410: focused-seeds/geometry/S9-spaces-limits-relative-glueing/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03KG: focused-seeds/geometry/S9-spaces-limits-relative-glueing/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 03KM: focused-seeds/geometry/S9-spaces-limits-relative-glueing/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $S = U \cup W$ be an open covering of a scheme. Then the functor*

$$FP_S \longrightarrow FP_U \times_{FP_{U\cap W}} FP_W$$

given by base change is an equivalence where FP_T is the category of algebraic spaces of finite presentation over the scheme T.

Proof. First, since $S = U \cup W$ is a Zariski covering, we see that the category of sheaves on $(Sch/S)_{fppf}$ is equivalent to the category of triples $(\mathcal{F}_U, \mathcal{F}_W, \varphi)$ where $\mathcal{F}_U$ is a sheaf on $(Sch/U)_{fppf}$, $\mathcal{F}_W$ is a sheaf on $(Sch/W)_{fppf}$, and

$$\varphi : \mathcal{F}_U|_{(Sch/U\cap W)_{fppf}} \longrightarrow \mathcal{F}_W|_{(Sch/U\cap W)_{fppf}}$$

is an isomorphism. See Sites, Lemma 04TS (note that no other gluing data are necessary because $U \times_S U = U$, $W \times_S W = W$ and that the cocycle condition is automatic for the same reason). Now, if the sheaf $\mathcal{F}$ on $(Sch/S)_{fppf}$ maps to $(\mathcal{F}_U, \mathcal{F}_W, \varphi)$ via this equivalence, then $\mathcal{F}$ is an algebraic space if and only if $\mathcal{F}_U$ and $\mathcal{F}_W$ are algebraic spaces. This follows immediately from Algebraic Spaces, Lemma 02WR as $\mathcal{F}_U \to \mathcal{F}$ and $\mathcal{F}_W \to \mathcal{F}$ are representable by open immersions and cover $\mathcal{F}$. Finally, in this case the algebraic space $\mathcal{F}$ is of finite presentation over S if and only if $\mathcal{F}_U$ is of finite presentation over U and $\mathcal{F}_W$ is of finite presentation over W by Morphisms of Spaces, Lemmas 03KG, 03KM, and 0410. □